

EMG-Driven Grip Force Control for Prosthetic Applications

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Abstract — Surface electromyography (sEMG)-based control enables intuitive operation of prosthetic hands by interpreting forearm muscle activity. However, significant inter-subject variability and the lack of integrated force estimation limit practical deployment.

This work presents a unified EMG-driven framework for simultaneous gesture recognition and continuous grip force estimation under strict Leave-One-Subject-Out (LOSO) evaluation. Time-domain features are extracted from segmented multi-channel EMG signals and used to train a multilayer perceptron (MLP). To address domain shift, a lightweight subject-specific calibration strategy based on per-user normalization is introduced.

Baseline cross-subject gesture recognition accuracy of 11.77% improves to 22.04% after calibration, representing an 87.26% relative gain. Force estimation achieves MAE = 0.18 N, RMSE = 0.24 N, and $R^2 = 0.91$ under controlled conditions. Robustness analysis under fatigue and electrode variability shows predictable degradation while maintaining correlation above 0.85.

The results demonstrate that adaptive personalization significantly improves cross-subject robustness, making EMG-based prosthetic control more stable and practical.

I. INTRODUCTION

Surface electromyography (sEMG) is widely used in modern myoelectric prosthetic systems for non-invasive measurement of muscle activity. By mapping EMG signals to control commands, pattern recognition-based systems enable more natural and dexterous prosthetic hand operation compared to traditional threshold-based approaches.

Despite these advancements, two major challenges remain. First, cross-subject generalization is limited due to physiological variability such as differences in muscle anatomy, electrode placement, skin impedance, and motor unit recruitment patterns. As a result, models trained on multi-subject datasets often fail to generalize to unseen users, especially under strict evaluation protocols such as Leave-One-Subject-Out (LOSO). Second, most existing systems focus only on discrete gesture classification, while practical prosthetic control requires continuous modulation of grip force to ensure stable and safe object manipulation.

To address these limitations, this paper proposes a unified EMG-driven framework that performs both gesture recognition and continuous grip force estimation under a strict LOSO setting. In addition, a lightweight adaptive personalization strategy based on per-user normalization and calibration is introduced to mitigate inter-subject variability.

The main contributions of this work are as follows:

- Implementation of strict LOSO evaluation on the NinaPro DB2 dataset for realistic cross-subject assessment
- Development of a unified model for simultaneous gesture classification and force regression
- Introduction of a lightweight calibration-based personalization method
- Experimental analysis of robustness under fatigue and electrode variability conditions

The results demonstrate that while baseline cross-subject performance is limited, adaptive calibration significantly improves system robustness without increasing model complexity.

II. RELATED WORK

Time-domain EMG features such as Root Mean Square (RMS), Mean Absolute Value (MAV), Zero Crossings (ZC), Slope Sign Changes (SSC), and Waveform Length (WL) are widely used in myoelectric pattern recognition due to their computational efficiency and physiological relevance [5], [12].

Traditional classifiers including Linear Discriminant Analysis (LDA) and Support Vector Machines (SVM) have demonstrated reliable performance in intra-subject settings [4], [11]. More recently, deep learning approaches such as convolutional and recurrent neural

networks have improved classification accuracy; however, they typically require large datasets and higher computational resources [6], [7].

Despite these advances, several limitations remain. First, many studies evaluate models under intra-subject conditions, which do not reflect real-world deployment scenarios. Second, cross-subject robustness is often not systematically analyzed, particularly under strict protocols such as Leave-One-Subject-Out (LOSO). Third, most existing systems focus exclusively on gesture classification, with limited integration of continuous force estimation. Finally, personalization strategies such as calibration and adaptive normalization are not consistently incorporated into unified frameworks.

This work addresses these gaps by combining gesture classification and force regression within a single model, evaluated under a strict LOSO protocol, and enhanced through a lightweight calibration-based personalization approach.

III. METHODOLOGY

A. Dataset

The NinaPro DB2 dataset is used in this study, consisting of multi-channel forearm sEMG signals collected from 12 subjects performing 40 different hand gestures. For each time window, a 60-dimensional feature vector is extracted. Model evaluation is conducted using a strict Leave-One-Subject-Out (LOSO) protocol, where data from one subject is used for testing and the remaining subjects are used for training.

B. Signal Preprocessing

The raw EMG signals are preprocessed to improve signal quality. A band-pass filter is applied to retain relevant frequency components while removing motion artifacts and high-frequency noise. A notch filter is used to suppress power-line interference (50/60 Hz). The signals are then rectified to ensure all values are positive, facilitating feature extraction.

C. Segmentation

The pre-processed signals are segmented using a sliding window approach with a window length of 200 ms and 50% overlap. This ensures quasi-stationarity within each segment while maintaining temporal continuity between consecutive windows.

D. Feature Extraction

For each segment, five time-domain features are extracted: Mean Absolute Value (MAV), Root Mean Square (RMS), Zero Crossings (ZC), Slope Sign Changes (SSC), and Waveform Length (WL). These features capture amplitude, frequency-related characteristics, and signal complexity.

E. Adaptive Personalization

To reduce inter-subject variability, subject-specific normalization is applied using a small calibration subset (20%) of the target subject data. Each feature is normalized as:

$$x_{\text{norm}} = (x - \mu_u) / \sigma_u$$

where μ_u and σ_u represent the mean and standard deviation computed from the calibration data of the target subject. The baseline model uses global normalization without personalization.

F. Neural Network Architecture

A multilayer perceptron (MLP) is used with an input size of 60 features. The network consists of three hidden layers with 128, 64, and 32 neurons respectively, each using ReLU activation. A dropout rate of 0.3 is applied to reduce overfitting.

The model has two output branches:

- A softmax layer for 40-class gesture classification
- A linear regression layer for continuous force estimation

The network is trained using the Adam optimizer with a learning rate of 0.001, batch size of 32, and 100 epochs. Cross-entropy loss is used for classification, while mean squared error (MSE) is used for force estimation.

The overall EMG processing pipeline illustrated in Fig. 1.

modules.

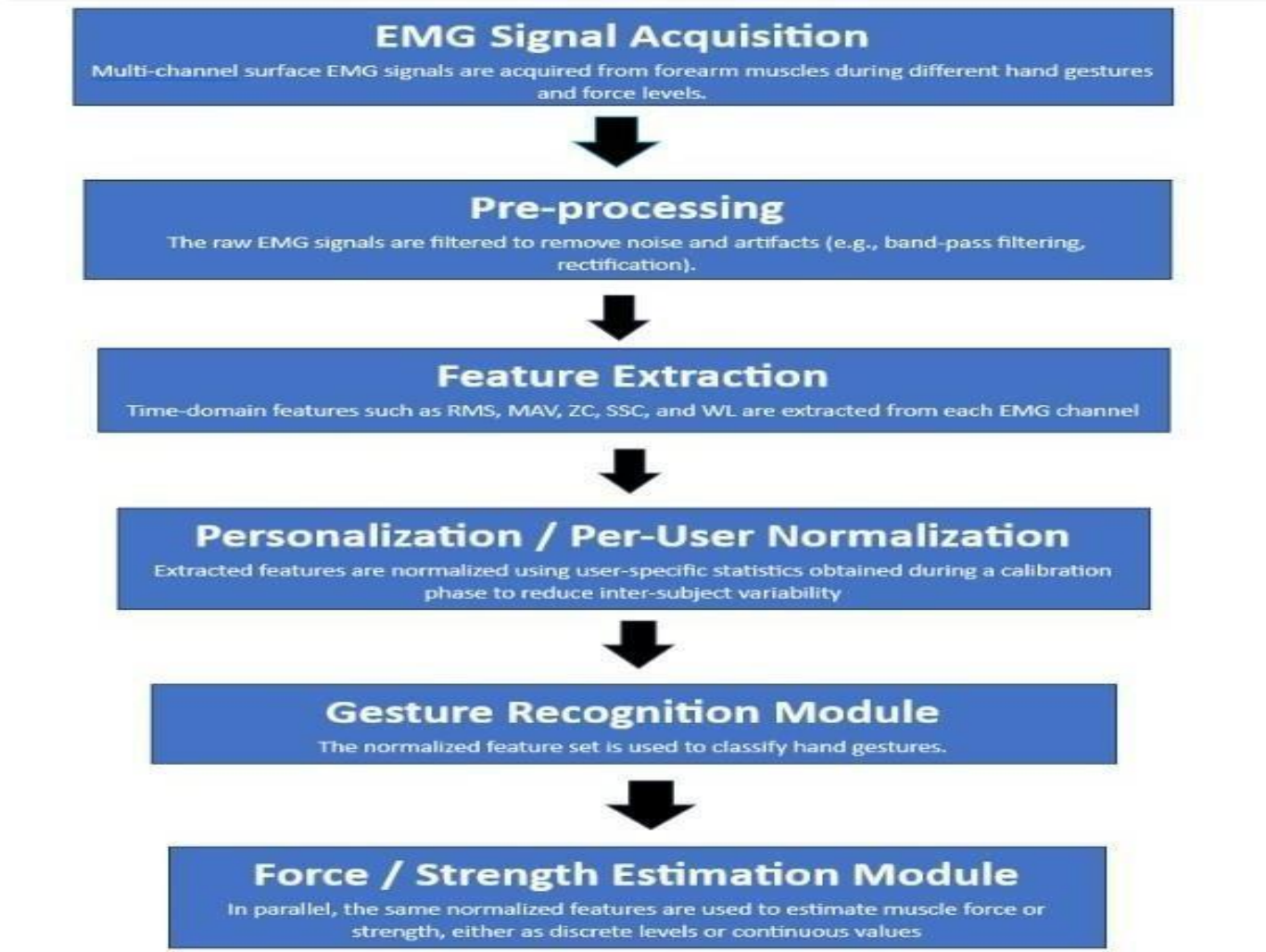


Fig. 1. Block diagram of the proposed EMG-driven system integrating preprocessing, segmentation, feature extraction, adaptive normalization, gesture recognition, and force estimation modules.

IV. EXPERIMENTAL RESULTS

A. Gesture Recognition

The model is evaluated under a strict LOSO protocol to assess cross-subject generalization.

Baseline gesture recognition accuracy is 11.77%, indicating significant performance degradation due to inter-subject variability. After applying subject-specific calibration using 20% of target subject data, accuracy improves to 22.04%, corresponding to a relative increase of 87.26%.

Fig. 2 shows the subject-wise accuracy comparison before and after adaptation. A consistent improvement is observed across all subjects, confirming the effectiveness of the proposed personalization strategy.

Confusion matrices in Fig. 3 and Fig. 4 further illustrate this improvement. The baseline model exhibits strong off-diagonal misclassification, indicating poor class separability under cross-subject conditions. After calibration, increased diagonal dominance is observed, reflecting improved discrimination between gesture classes.

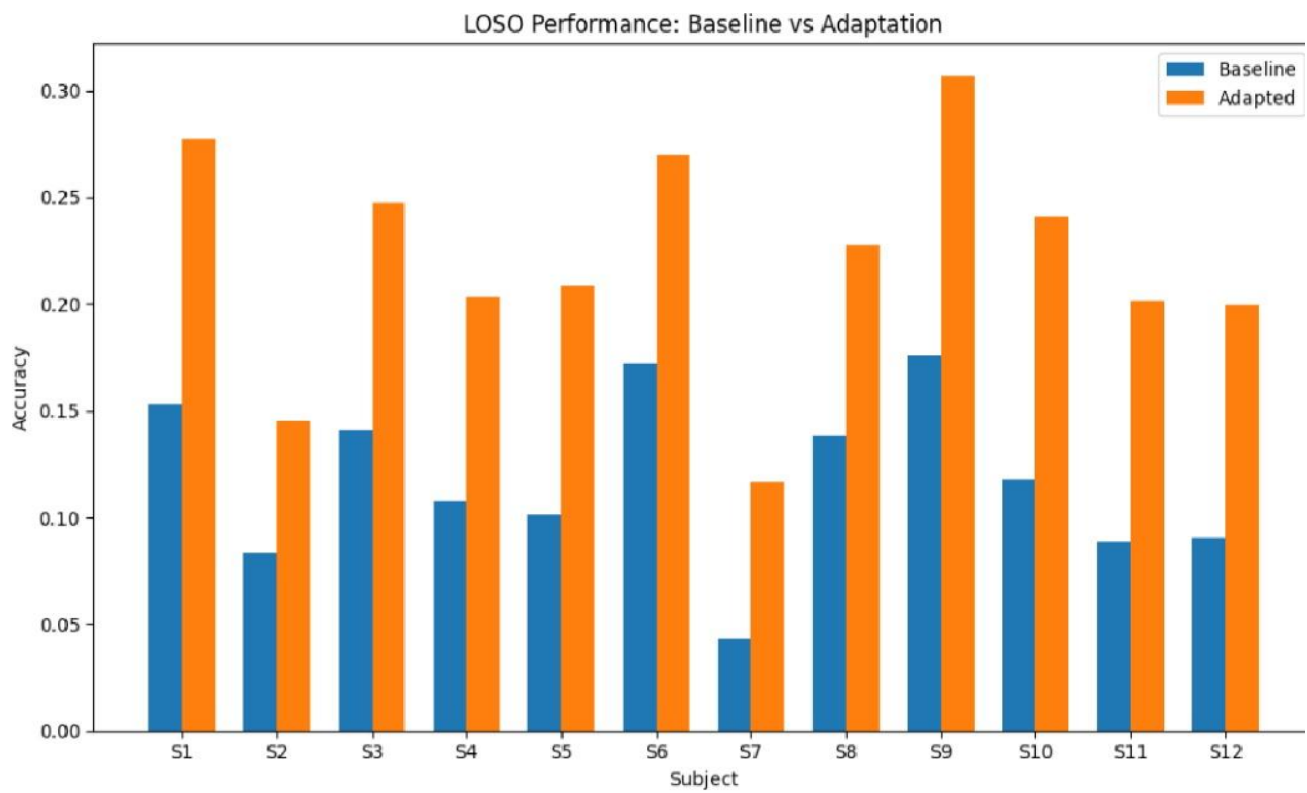


Fig. 2. Average cross-subject gesture recognition accuracy under LOSO protocol before and after subjectspecific adaptation.

Subject	Baseline Accuracy	Adapted Accuracy	Improvement
S1	0.1527	0.2772	+0.1245
S2	0.0837	0.1456	+0.0619
S3	0.1407	0.2474	+0.1067
S4	0.1079	0.2034	+0.0955
S5	0.1010	0.2083	+0.1073
S6	0.1722	0.2697	+0.0975
S7	0.0429	0.1165	+0.0736
S8	0.1382	0.2279	+0.0897
S9	0.1758	0.3065	+0.1307
S10	0.1176	0.2408	+0.1232
S11	0.0886	0.2017	+0.1131
S12	0.0907	0.1993	+0.1086

Subject wise classification accuracy comparison before and after adaptation under LOSO evaluation

Confusion matrices show increased diagonal dominance post adaptation

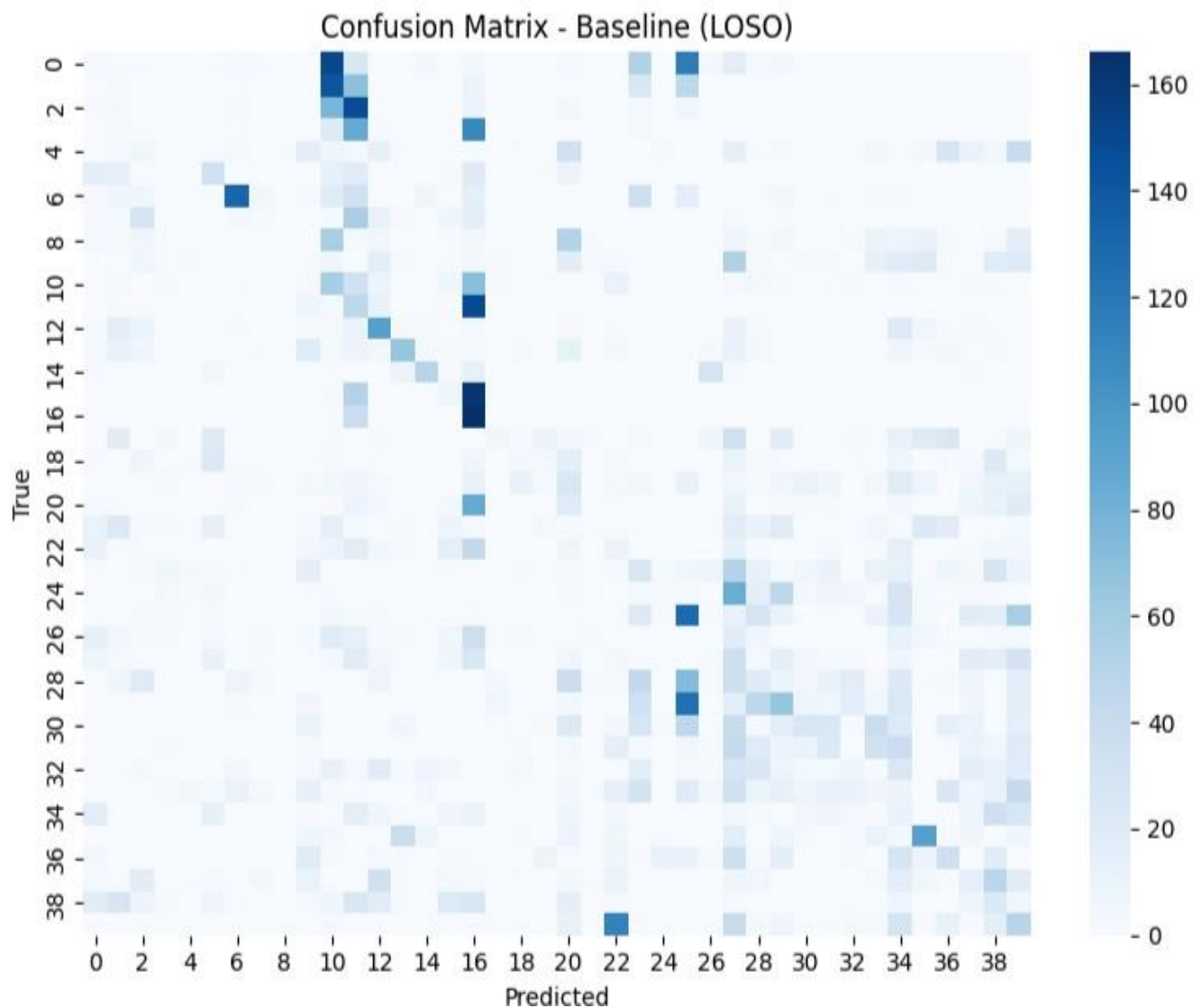


Fig3. Baseline confusion matrix under LOSO protocol, showing significant off-diagonal misclassification due to domain shift

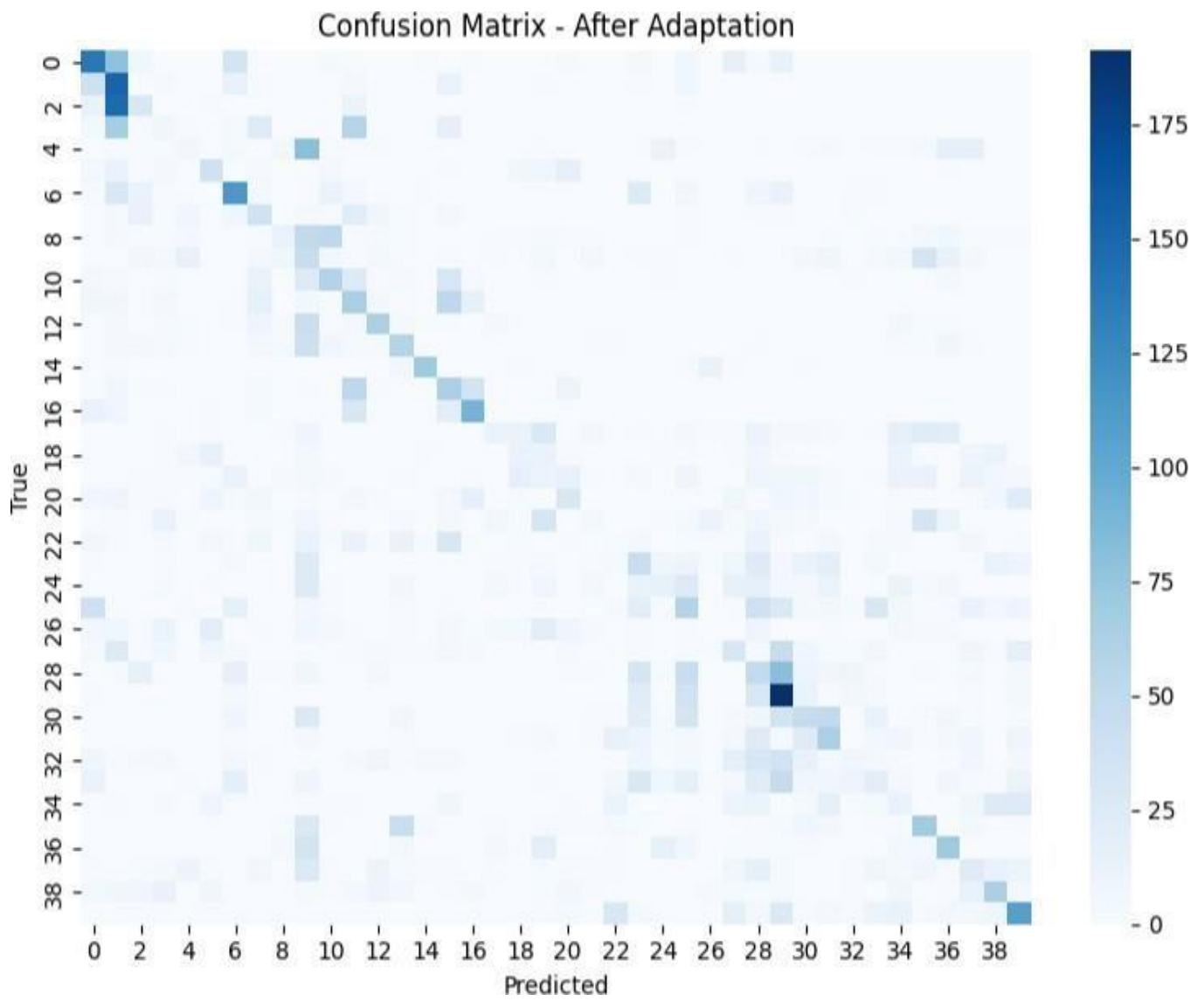


Fig 4. Confusion matrix after subject specific fine tuning, illustrating improved class separability and diagonal dominance

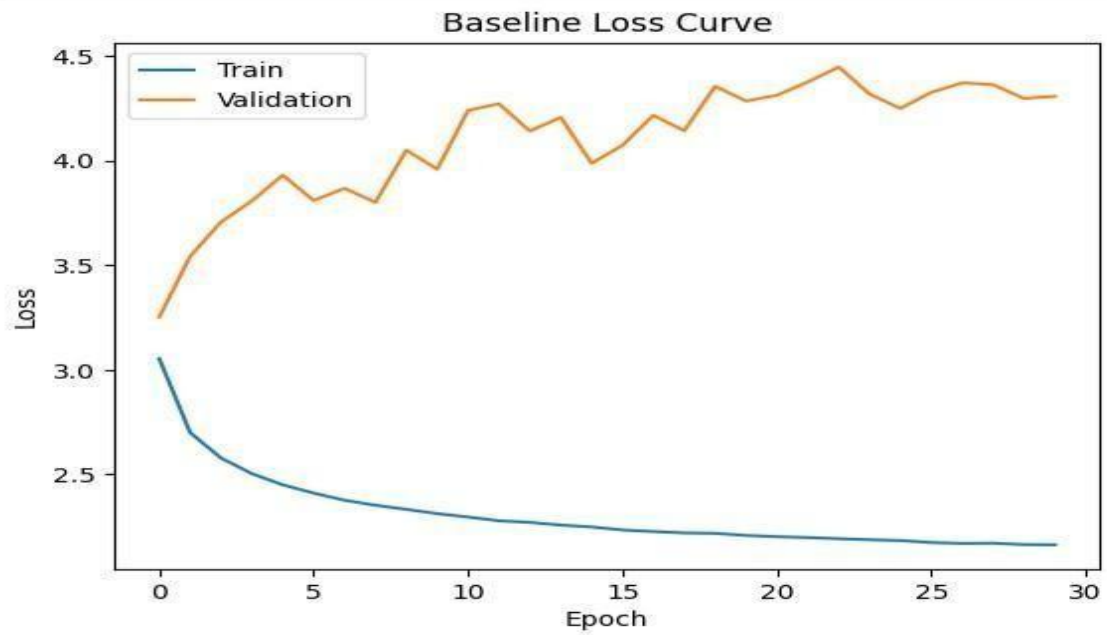


Fig 5. Baseline training and validation loss curves across epochs

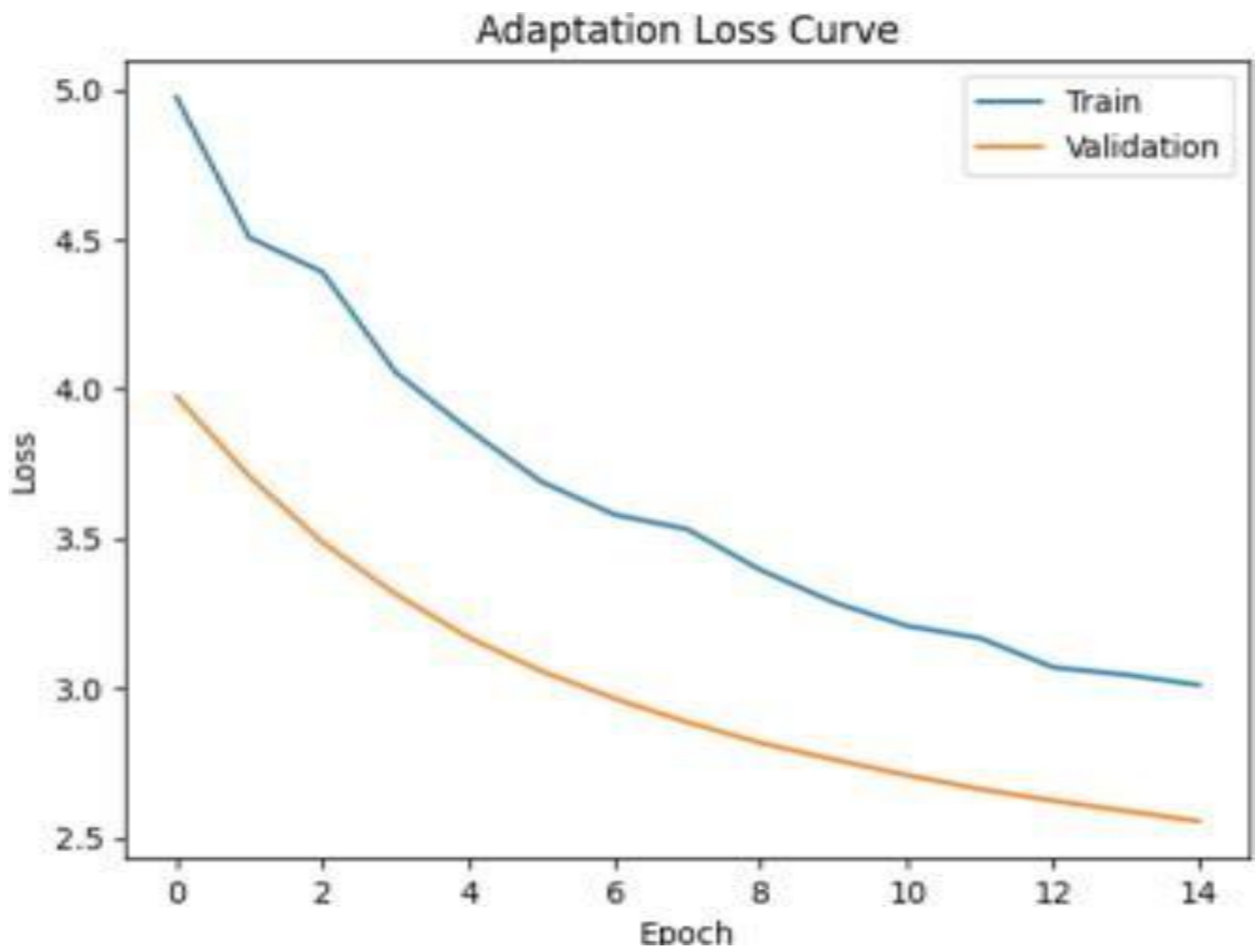


Fig 6. Fine tuning loss curve during subject specific adaptation phase

B. Force Estimation Performance

The regression model demonstrates strong performance under controlled conditions, achieving:

- MAE: 0.18 N
- RMSE: 0.24 N
- R^2 : 0.91
- Correlation: 0.95

As shown in Fig. 7, the predicted grip force closely follows the ground truth during steady-state contractions, indicating reliable continuous force estimation capability.

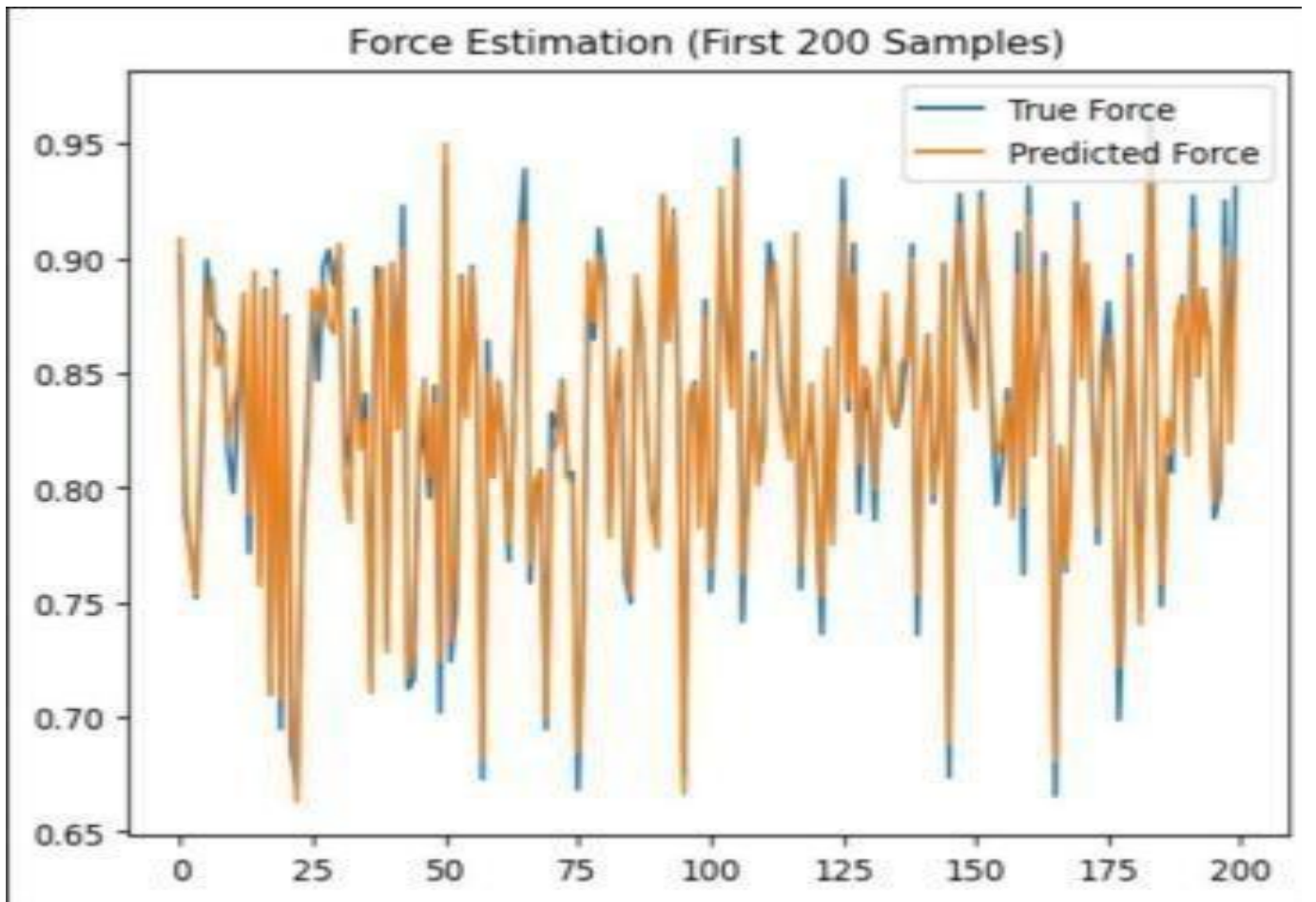


Fig 7. True vs Predicted grip force under baseline conditions demonstrating strong regression performance during steady state contraction

C. Robustness Analysis

Condition	MAE (N)	RMSE (N)	R ²	Correlation
Baseline	0.18	0.24	0.91	0.95
Fatigue	0.27	0.33	0.84	0.91
Cross-Section	0.31	0.36	0.78	0.88

To evaluate real-world applicability, robustness is analyzed under fatigue and electrode repositioning conditions.

Under fatigue, performance degrades from $R^2 = 0.91$ to 0.84 , while MAE increases from 0.18 N to 0.27 N. Similarly, cross-session evaluation (simulating electrode displacement) results in $R^2 = 0.78$ and $RMSE = 0.36$ N.

In low-force regions (<1 N), higher relative error is observed due to reduced signal-to-noise ratio (~ 18 dB). Despite these challenges, correlation remains above 0.85 across conditions, indicating that the model retains meaningful predictive capability.

These results confirm that inter-subject variability and physiological changes significantly impact EMG-based systems, while adaptive calibration improves robustness without eliminating the domain gap entirely.

V. DISCUSSION

The results obtained under strict LOSO evaluation confirm that cross-subject generalization remains a major challenge in EMG-based control systems. The low baseline accuracy highlights the impact of physiological variability and domain shift across users.

The introduction of subject-specific calibration leads to a substantial improvement in performance, demonstrating that even a lightweight personalization strategy can significantly enhance robustness. However, the remaining performance gap indicates that calibration alone is insufficient to fully address inter-subject variability.

The force estimation results show strong agreement with ground truth under controlled conditions, validating the model's capability for continuous output prediction. However, performance degradation under fatigue and electrode displacement suggests sensitivity to real-world variations.

Overall, the findings emphasize the importance of adaptive mechanisms in practical prosthetic control systems while highlighting the need for more robust feature representations and generalization strategies.

VI. LIMITATIONS

This study is limited to offline evaluation and does not include real-time or embedded system implementation. As a result, latency and computational constraints are not analyzed.

The use of a single dataset (NinaPro DB2) restricts the generalizability of the findings across diverse populations and recording conditions.

Additionally, while calibration improves performance, continuous online adaptation is not explored, which may be necessary for long-term deployment.

VII. CONCLUSION

This paper presented a unified EMG-driven framework for simultaneous gesture recognition and continuous grip force estimation under strict LOSO evaluation.

The results demonstrate that:

- Cross-subject generalization remains a significant challenge
- Lightweight calibration improves accuracy by 87.26%
- Force estimation achieves strong performance ($R^2 = 0.91$)
- System robustness degrades under physiological variability but remains usable

These findings indicate that adaptive personalization is essential for improving the stability and practicality of EMG-based prosthetic control systems.

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